Bridges: the one-page version

What the activity is

Each drawing is a town: white circles are islands, grey bands are bridges. A counter sits on every bridge; a token walks from island to island, and crossing a bridge means picking up its counter, so no bridge can be used twice. The question all week: can you pick up every counter, and which island do you have to start on?

Materials (whole room; about 20 minutes to gather and test)

- Counters about 1 inch or smaller (about 150 in all; K–1 needs 20 per pair). Tokens that look different from the counters, 2 per pair (one walks, one marks the start).
- Yellow pattern-block hexagons (about 40) and 4.5-inch craft sticks (about 60) for the building problems (K–1 P7, Grades 2–3 P8). Two sticks side by side make two bridges between the same pair of islands.
- Coloured pencils, pencils, scrap paper; small sticky notes to letter hexagons (Grades 2–3). Masking tape for the optional floor town.
- Print single-sided at 100%, not "fit to page", or the bridges shrink below counter size. Before the session, check that counters fit the tightest town (K–1 page 9, second town).

Launch (two minutes, whole group)

Use a house-shaped town (five islands, six bridges) taped on the floor or K–1 page 1. "Every bridge has a counter. Crossing a bridge, you pick up its counter; a bridge with no counter is closed." A child starts on a bottom corner, walks, and gets stuck with counters left. "Would a different start pick up every counter?" Don't say which start works.

How it runs, and what Problem 1 is for

- **Every band:** put every counter back before each new try. Two counters stacked on one bridge means it can be crossed twice (take the top one). Each tracing picture is printed twice for a second try.
- **Problem 1** is a few small towns to walk: K–1 colours the island it started on, and Grades 2–3 and 4–5 write the walk as letters, like A B C A. Its job is to get the walking rule into hands; children find that some starts work and some get stuck.
- Core of the hour: K–1 Problems 1–3, Grades 2–3 Problems 1–4, Grades 4–5 Problems 1–5. The rest is for children who get further.

The surprise (don't tell it; let the counters show it)

Count the bridges at each island. If every island has an even number, start anywhere and you end where you started. If exactly two islands are odd, start at one and you end at the other. If more are odd, it can't be done: every time you pass through an island you use its bridges two at a time, so an odd island must be the start or the end.

Key answers

- **K–1.** P2: two squares, only the two middle islands; big triangle, any island. P3: X, yes, yes, yes, X (the three-arm star fails: three tips, each with one bridge). P4: yes, X, X, yes, yes. P5: cross out the envelope and the circle with a plus. P6: join any two islands that have 1 or 3 bridges. P8: arc town all 6 bridges; square with tails, the 2 tails; square with centre, the 4 sides.
- **Grades 2–3.** P1: house from B or C. P2: town 1 none, town 2 A or D, town 3 D or F, town 4 anywhere. P6 is the rule above. P9: impossible; walk it backwards, or round its loop from another island, to get a second start.
- **Grades 4–5.** P4 is the rule and why. P5, Königsberg: impossible, since all four land areas are odd; one new bridge between any two areas makes a walk possible, and two new bridges make one that ends where it started. P6: a town never has exactly one odd island, since the bridge counts add up to twice the number of bridges.